

FedBaS: Bandwidth-Aware Feature Selection Strategies for Efficient Federated Distillation

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Abstract—Federated learning (FL) allows collaborative model training without data centralization, but its scalability is usually limited by communication overhead. Although Federated Averaging (FedAvg) communicates the full model parameters, federated distillation (FD) can reduce the communication cost by transmitting knowledge representations; yet, the accuracy-efficiency tradeoffs of FD are still not well understood. We propose FedBaS, a bandwidth-efficient feature selection framework for federated distillation. By a systematic evaluation, we demonstrate that FD can reach accuracy comparable to FedAvg with a greatly reduced communication cost. We also examine logit-based and feature-based distillation methods and show that feature distillation accelerates convergence and enhances robustness but at the cost of increased bandwidth. To mitigate this cost, FedBaS proposes novel variance-, weight-, and mutual information-based feature selection methods with greatly reduced communication cost while maintaining accuracy improvements. Our study offers valuable guidance for implementing communication-efficient federated learning in bandwidth-limited mobile edge and future wireless communication scenarios, such as 6G.

Index Terms—Bandwidth efficiency, edge cloud, peer-to-peer networks, federated learning, feature selection, knowledge distillation.

I. INTRODUCTION

Federated learning (FL) has become a prominent paradigm for distributed model training in scenarios where raw data cannot be centralized due to privacy, security, or regulatory concerns. Instead of transmitting local datasets, participating clients train models locally and exchange information with a coordinating server. While this paradigm preserves data locality, it comes at the cost of *high communication overhead*, as conventional approaches such as Federated Averaging (FedAvg) [1] require the transmission of complete model parameters at every training round. For large deep neural networks, this process is bandwidth-intensive and may render FL impractical in edge environments characterized by limited network capacity and heterogeneous devices.

To mitigate communication bottlenecks, recent work has explored *federated distillation (FD)* as an alternative to parameter averaging [2], [3]. In FD, clients do not share raw parameters but instead transmit intermediate knowledge representations such as output logits or feature embeddings. This significantly reduces the amount of information exchanged and provides a pathway toward communication-efficient FL. However, despite its promise, the *tradeoffs between accuracy and bandwidth consumption in FD remain insufficiently characterized*. Existing literature often reports that while FD reduces overhead compared to FedAvg, it may suffer from

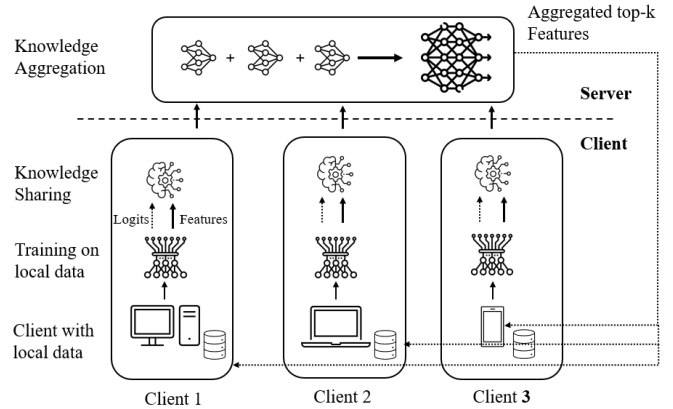


Fig. 1. FedBaS workflow: clients train locally, share logits and selected features, and the server aggregates knowledge for redistribution.

degraded accuracy, particularly when relying solely on logit sharing [4]. Conversely, feature-based distillation transmits richer intermediate signals that can improve convergence and accuracy [5], but at a substantial communication cost.

This tension highlights a core challenge: *how can we preserve the accuracy benefits of feature distillation while alleviating its bandwidth burden?* This challenge is particularly relevant in emerging wireless edge systems, where clients operate under fluctuating link quality, asymmetric uplink–downlink capacity, and strict energy constraints. In such settings, communication efficiency is not only a performance concern but a system-level design requirement. Consequently, understanding *what* knowledge to transmit—and how much—is as important as *how often* communication occurs. Addressing this question is crucial for real-world deployments of FL at the network edge, where bandwidth limitations are often the primary bottleneck. Moreover, little systematic work has been conducted to analyze different feature selection strategies that could make feature sharing more bandwidth-efficient without sacrificing model performance.

In this paper, we propose *FedBaS* (Federated Bandwidth-aware Selection), a framework designed to explore and optimize the tradeoffs in communication-efficient federated distillation. Our study is organized into three parts. First, we establish a comparative baseline by contrasting FedAvg with federated distillation, demonstrating that FD can achieve accuracy on par with FedAvg while requiring less communication. Second, we analyze the distinction between logit-based and feature-based distillation, showing that while logits provide

communication savings, feature-based sharing yields stronger performance improvements. Third, we introduce and evaluate three feature selection strategies—variance-based, weight-based, and mutual information-based—that significantly reduce the communication cost of feature distillation while retaining its performance gains.

Our contributions are summarized as follows:

- *Communication–accuracy tradeoff analysis:* We conduct a systematic comparison of FedAvg and federated distillation, quantifying the bandwidth savings of FD while showing its competitive accuracy.
- *Logit vs. feature distillation comparison:* We evaluate the accuracy–bandwidth tradeoffs of logit and feature sharing, highlighting the conditions under which feature distillation provides clear benefits.
- *Bandwidth-aware feature selection:* We propose FedBaS, a set of three feature selection strategies (variance-based, weight-based, and mutual information-based) that reduce the communication cost of feature distillation without compromising its accuracy.

In this way, *FedBaS* offers a useful and sound solution to the problem of bandwidth-efficient federated learning. In addition to the above improvements in communication efficiency, our findings also offer design insights on *what* knowledge should be communicated in bandwidth-limited scenarios. Such findings can be directly applied to the design of FL systems in bandwidth-limited environments such as mobile and wireless networks, edge computing architectures, and IoT networks.

II. RELATED WORK

Communication-efficient federated learning. A central bottleneck in FL is the per-round uplink/downlink payload of model parameters. Early work explored client-side local updates and server averaging (FedAvg) to amortize communication over multiple SGD steps [1], followed by communication compression including quantization, sparsification, and sketching. Representative techniques include QSGD [6], TernGrad [7], top- k sparsification [8], deep gradient compression (DGC) [9], and error-feedback/1-bit SGD. System-level strategies leverage periodic aggregation and partial participation [10], update clipping and momentum correction [9], or server-side variance reduction (e.g., SCAFFOLD) to stabilize non-IID training with fewer rounds [11]. More recent directions include adaptive communication frequency and quantized aggregation (e.g., FedPAQ) [12], showing that careful protocol design can lower bytes-per-accuracy without modifying local models. Our work is orthogonal: instead of compressing *parameters*, we reduce the payload by transmitting *knowledge representations* and then selectively sub-sampling features under a bandwidth budget.

Knowledge distillation (KD) and federated distillation (FD). KD transfers information from teacher to student using soft targets or intermediate representations [13], [14]. In FL, FD replaces parameter exchange with *logits* or *features* evaluated on a proxy/public set. Pioneering FD frameworks include

FedMD [15] and variants that ensemble client predictions to build server-side teachers [4], as well as data-free and proxy-data approaches (e.g., FedDF) that avoid sharing raw data while maintaining utility [16]. Other works study distillation on heterogeneous client architectures [15], cross-domain or non-IID regimes via calibration/temperature scaling [3], and robustness to client drift by re-weighting soft targets or mixing pseudo-labels with local ground-truth [3]. While logit-sharing is extremely communication-light (scales with the number of classes), feature-based distillation tends to improve convergence by providing richer intermediate supervision [14], at the cost of larger payloads. *FedBaS* addresses this gap by selecting an informative subset of features, preserving accuracy benefits while reducing bytes.

Feature selection and mutual information. Classical filter methods rank features by variance or correlation, while wrapper/embedded methods couple selection with model training. Mutual information (MI) has long been used to pick features with the greatest predictive content, e.g., Battiti’s mutual information-based feature selection and mRMR’s redundancy-aware criterion [17], [18]. Information-theoretic views of representation learning—including the information bottleneck and variational IB—justify keeping features that maximize $I(Y; H)$ under a complexity (rate) constraint [19], [20]. In deep KD, MI-inspired objectives can encourage students to preserve teacher structure at intermediate layers [14]. Our selection criteria operationalize these ideas with *variance-based*, *weight-based*, and *MI-based* rules that are inexpensive to compute on-device and maps onto a bandwidth budget.

Personalization and non-IID robustness. FL under client heterogeneity motivates personalization layers/heads (FedPer), proximal regularization (FedProx), and bilevel/meta-learning formulations [21]–[23]. KD has also been used as a *personalization bridge* by letting clients distill global knowledge into local models without strict parameter alignment [15], [16]. Our study is complementary: we focus on the *communication–accuracy* frontier of distillation itself (logits vs. features) and show that selective feature sharing can retain accuracy gains even under bandwidth limits, providing a knob that can be combined with the above personalization methods.

Positioning of FedBaS. Where compression methods shrink *gradients/parameters* and standard FD shrinks *semantics* (logits), *FedBaS* targets the middle ground: it keeps semantically rich *features* but sends only the most informative subset. This offers a practical path to close the known gap between the efficiency of logit-based FD and the accuracy of feature-based FD, especially in edge settings where uplink is scarce.

III. SYSTEM MODEL AND OBJECTIVES

We consider a federated learning (FL) system with N clients $\{C_1, C_2, \dots, C_N\}$ and a central server. Each client C_i owns a local dataset $\mathcal{D}_i$ that remains private. The global objective is $F(\mathbf{w}) = \sum_{i=1}^N p_i f_i(\mathbf{w})$,

where $f_i(\mathbf{w})$ is the local loss function of client i , $\mathbf{w}$ denotes the model parameters, and $|\mathcal{D}| = \sum_{i=1}^N |\mathcal{D}_i|$.

TABLE I
SYMBOLS AND NOTATIONS

Symbol	Description
N	Number of clients in the federated learning system
C_i	i -th client participating in training
$\mathcal{D}_i$	Local dataset of client C_i
$\mathbf{w}$	Global model parameters
$\mathbf{w}_i$	Local model parameters of client C_i
$f_i(\mathbf{w})$	Local loss function of client i
$F(\mathbf{w})$	Global objective function (weighted average of local risks)
$\mathbf{z}_i$	Logit (prediction vector) produced by client C_i
T	Temperature parameter in logit distillation loss
$\mathbf{h}_i^l$	Feature representation of client i at layer l
$\tilde{\mathbf{h}}_i^l$	Selected subset of features after applying operator $\mathcal{S}$
$\bar{\mathbf{z}}$	Aggregated average of logits across clients
$\bar{\mathbf{h}}^l$	Aggregated feature map across clients at layer l
$\mathcal{S}(\cdot; \Theta)$	Feature selection operator with criterion Θ
$\mathcal{C}_{\text{FedAvg}}$	Per-round communication cost of FedAvg
$\mathcal{C}_{\text{FD-logit}}$	Per-round communication cost of FD-logit
$\mathcal{C}_{\text{FD-feature}}$	Per-round communication cost of FD-feature

A. Communication in FedAvg

In Federated Averaging (FedAvg) [1], each training round requires all clients to upload their local parameters $\mathbf{w}_i \in \mathbb{R}^d$ and download the aggregated global model $\mathbf{w}$. The per-round communication cost is

$$\mathcal{C}_{\text{FedAvg}} = N \cdot (|\mathbf{w}|_{\text{up}} + |\mathbf{w}|_{\text{down}}), \quad (1)$$

where $|\mathbf{w}|$ denotes the number of model parameters. For modern deep networks, this overhead is often the dominant bottleneck.

B. Federated Distillation

Federated distillation (FD) [2], [3] reduces communication cost by transmitting intermediate knowledge representations instead of full model parameters. Two common variants are considered.

Logit-based distillation: Clients transmit prediction vectors $\mathbf{z}_i = f_{\mathbf{w}_i}(x)$ evaluated on public inputs $x \in \mathcal{B}$. The server aggregates these predictions to form soft targets, and clients are trained using the temperature-scaled KL divergence

$$\mathcal{L}_{\text{logit}} = \sum_{x \in \mathcal{B}} \text{KL}(\mathbf{z}_i/T \parallel \bar{\mathbf{z}}/T), \quad (2)$$

where T is the distillation temperature and $\bar{\mathbf{z}}$ denotes the average prediction across clients.

Feature-based distillation: Clients instead transmit intermediate representations $\mathbf{h}_i^l$ from a selected layer l . Knowledge transfer is achieved by minimizing a feature-matching loss

$$\mathcal{L}_{\text{feat}} = \sum_l \|\mathbf{h}_i^l - \bar{\mathbf{h}}^l\|_2^2, \quad (3)$$

where $\bar{\mathbf{h}}^l$ is the aggregated feature representation.

Logit sharing is typically more bandwidth-efficient, while feature sharing provides richer supervision at higher communication cost. Whether this added cost translates into improved accuracy is an empirical question, which we investigate in this work.

C. Bandwidth-Aware Feature Selection

Full feature maps can be large, making feature-based distillation bandwidth-intensive, with per-round cost $\mathcal{C}_{\text{feat}} = \sum_{l \in \mathcal{L}} |\mathbf{h}_i^l|$. To reduce this overhead, we introduce *FedBaS*, a framework for bandwidth-aware feature selection. Instead of transmitting all features, each client selects a subset:

$$\tilde{\mathbf{h}}_i^l = \mathcal{S}(\mathbf{h}_i^l; \Theta), \quad (4)$$

where $\mathcal{S}(\cdot)$ is a selection operator defined by criterion Θ .

In this paper, we study three criteria for Θ :

- 1) Variance-based selection: prioritizing features with high activation variance.
- 2) Weight-based selection: selecting features with large associated weight magnitudes.
- 3) Mutual information (MI)-based selection: choosing features that maximize MI with labels.

Our objective is to investigate the tradeoff between accuracy and communication cost. Formally, the goal is to

$$\min_S \mathcal{C}_{\text{feat}} \quad \text{s.t.} \quad \Delta \text{Acc} \leq \epsilon, \quad (5)$$

where ΔAcc is the accuracy difference relative to a baseline (e.g., FedAvg or full feature sharing) and ϵ is a tolerable threshold.

IV. ACCURACY UNDER BANDWIDTH BUDGETS

We analyze the communication–accuracy tradeoffs in federated distillation by characterizing how bandwidth scales across methods, how feature selection affects label information, and how these effects align with classical rate–distortion behavior.

A. Communication Scaling

Let d be the number of model parameters, C the number of classes, d_l the feature dimension at layer l , N the number of clients, and R the number of rounds. The per-round communication scales as $\mathcal{O}(d)$ for FedAvg, $\mathcal{O}(C)$ for logit-based distillation, $\mathcal{O}(d_l)$ for full feature-based distillation, and $\mathcal{O}(k)$ for top- k feature sharing. Over R rounds with N clients, total communication scales as $\mathcal{O}(RNd)$, $\mathcal{O}(RNC)$, $\mathcal{O}(RNd_l)$, and $\mathcal{O}(RNk)$, respectively. This highlights that FedAvg becomes prohibitive as model size grows, while feature-based methods allow tunable cost via layer choice and feature selection.

B. Information-Theoretic View of Feature Selection

Feature selection reduces communication by transmitting a subset $\tilde{\mathbf{h}}^l = \mathcal{S}(\mathbf{h}^l)$ of the full representation $\mathbf{h}^l$. Since $\tilde{\mathbf{h}}^l$ is a deterministic function of $\mathbf{h}^l$, the data-processing inequality gives $I(Y; \tilde{\mathbf{h}}^l) \leq I(Y; \mathbf{h}^l)$. The resulting accuracy loss is therefore controlled by the mutual-information gap $I(Y; \mathbf{h}^l) - I(Y; \tilde{\mathbf{h}}^l)$, implying that accuracy can be preserved when the selected features retain most label-relevant information. This motivates ranking features by variance, classifier weight magnitude, or mutual-information proxies.

Algorithm 1 FedBaS: One Communication Round

Input: Clients $\{C_i\}_{i=1}^N$, local datasets $\{\mathcal{D}_i\}$, selection rule $\mathcal{S}(\cdot; \Theta)$, budget B , temperature T

Output: Updated local models $\{\mathbf{w}_i\}$

- 1: Server broadcasts minibatch $\mathcal{B}$ and metadata $(\mathcal{L}, B, \Theta)$
 - 2: **for all** clients C_i
 - 3: Train locally on $\mathcal{D}_i$ to update $\mathbf{w}_i$
 - 4: Compute logits $\mathbf{z}_i(x)$ and features $\mathbf{h}_i^l(x)$ for $x \in \mathcal{B}$
 - 5: Select features $\tilde{\mathbf{h}}_i^l(x) \leftarrow \mathcal{S}(\mathbf{h}_i^l(x); \Theta)$
 - 6: Upload $\mathbf{z}_i(x)$ and $\tilde{\mathbf{h}}_i^l(x)$ to server
 - 7: Server aggregates $\bar{\mathbf{z}}(x) = \frac{1}{N} \sum_i \mathbf{z}_i(x)$ and $\bar{\mathbf{h}}^l(x) = \frac{1}{N} \sum_i \tilde{\mathbf{h}}_i^l(x)$
 - 8: Server broadcasts $\bar{\mathbf{z}}(x)$ and $\bar{\mathbf{h}}^l(x)$
 - 9: **for all** clients C_i
 - 10: Compute distillation losses $\mathcal{L}_{\text{logit}}, \mathcal{L}_{\text{feat}}$
 - 11: Update $\mathbf{w}_i$ with SGD on $\mathcal{D}_i$ plus distillation losses
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C. Rate–Distortion Perspective

The feature-selection problem can be viewed as minimizing accuracy loss under a communication constraint: $D(R) = \min_{\mathcal{C}(\mathcal{S}) \leq R} \mathbb{E}[\Delta \text{Acc}(\mathcal{S})]$. As the feasible set expands with rate R , achievable accuracy improves monotonically, typically with diminishing returns. Early bits capture the most informative structure, while additional bandwidth refines details—analogueous to progressive video compression.

D. Implications for FedBaS

Together, these insights predict a concave accuracy–bandwidth frontier in which most gains arise from a small number of informative features. FedBaS operationalizes this principle by enforcing per-round bandwidth budgets via top- k feature selection, fp16 transmission, capped public sets, and periodic head-only broadcasts. By concentrating communication on high-value coordinates, FedBaS achieves the accuracy-per-megabyte improvements observed empirically.

V. FEDBAS FRAMEWORK

In this section, we present *FedBaS* (Federated Bandwidth-aware Selection), a framework that unifies traditional federated learning, federated distillation, and feature selection into a single communication–accuracy optimization pipeline. FedBaS generalizes existing strategies and extends them by incorporating bandwidth-aware feature selection. FedBaS is designed to explore three key questions: (i) How does federated distillation compare to FedAvg in terms of bandwidth usage and accuracy? (ii) What are the tradeoffs between logit-based and feature-based distillation? (iii) Can we selectively transmit features to reduce communication cost while maintaining performance?

A. Overall Workflow

Each communication round in FedBaS follows the procedure outlined in Algorithm 1, proceeding in three main stages:

- 1) *Local training:* Each client C_i updates its parameters $\mathbf{w}_i$ on its local dataset $\mathcal{D}_i$ using stochastic gradient

descent or another optimizer. This corresponds to the local update step in Algorithm 1 (lines 2–3).

- 2) *Knowledge sharing:* Instead of transmitting $\mathbf{w}_i$ directly, clients compute logits $\mathbf{z}_i$ and feature maps $\mathbf{h}_i^l$, then apply the feature selection operator $\mathcal{S}$ to obtain $\tilde{\mathbf{h}}_i^l$. The selected representations, together with logits, are uploaded to the server (lines 4–6).
- 3) *Aggregation and redistribution:* The server aggregates the received logits and features into $\bar{\mathbf{z}}$ and $\bar{\mathbf{h}}^l$, broadcasts them back, and clients update their models by combining standard SGD with distillation losses (lines 7–11).

This structure highlights that FedAvg, FD-logit, and FD-feature emerge as special cases of Algorithm 1. FedAvg corresponds to transmitting the full parameter vector $\mathbf{w}_i$, FD-logit corresponds to sending $\mathbf{z}_i$, and FD-feature corresponds to sending $\mathbf{h}_i^l$. FedBaS generalizes further by applying $\mathcal{S}$ to $\mathbf{h}_i^l$, yielding bandwidth-aware variants under the same unified workflow.

B. Step 1: FedAvg vs. Federated Distillation

The first component of FedBaS evaluates the communication–accuracy tradeoff between FedAvg and federated distillation (FD). FedAvg exchanges full model parameters, incurring high communication cost, whereas FD transmits compact knowledge representations such as logits or features. We quantify efficiency as $\text{Efficiency}(M) = \text{Accuracy}(M)/\mathcal{C}(M)$, where $\mathcal{C}(M)$ denotes the communication cost of method M . This formulation enables a direct comparison of whether FD achieves comparable accuracy to FedAvg under significantly lower bandwidth, which is critical for bandwidth-limited wireless and edge deployments.

C. Step 2: Logit vs. Feature Distillation

The second component of FedBaS contrasts logit-based and feature-based federated distillation (FD). In logit-based FD, each client transmits class logits $\mathbf{z}_i \in \mathbb{R}^C$, incurring a small, model-size-independent communication cost, but potentially losing intermediate representational information. Feature-based FD instead exchanges intermediate representations $\mathbf{h}_i^l \in \mathbb{R}^{d_l}$ from layer l , which are typically much larger ($d_l \gg C$) but encode richer task-specific structure. FedBaS evaluates whether the additional bandwidth required for feature transmission yields measurable accuracy improvements over logit-only distillation.

D. Step 3: Bandwidth-Aware Feature Selection

The third component of FedBaS reduces the communication cost of feature-based distillation by selectively transmitting only informative feature coordinates. Specifically, each client applies a selection operator $\tilde{\mathbf{h}}_i^l = \mathcal{S}(\mathbf{h}_i^l; \Theta)$, where $\mathcal{S}$ retains a subset of features according to criterion Θ . We consider variance-based selection (high activation variance), weight-based selection (large classifier weights), and mutual-information-based selection (high predictive relevance). The resulting objective is to maximize accuracy under a bandwidth

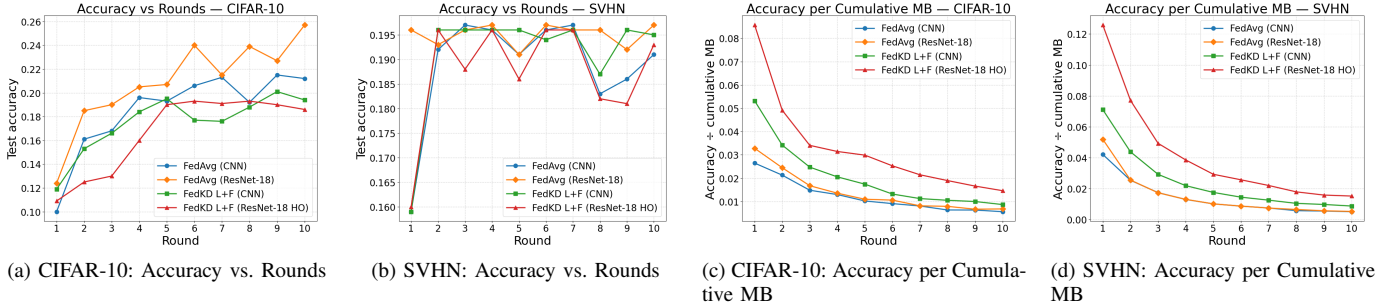


Fig. 2. FedAvg vs. FedKD (logits+features) on CIFAR-10 and SVHN. Panels (a)–(b): accuracy over rounds. Panels (c)–(d): accuracy normalized by cumulative communication (MB).

constraint, i.e., $\max_{\mathcal{S}} \text{Accuracy}(\mathcal{S})$ s.t. $\mathcal{C}(\mathcal{S}) \leq B$, which explicitly characterizes the bandwidth–accuracy tradeoff explored in our experiments.

E. Framework Contributions

FedBaS provides a unified pipeline that enables systematic comparison of federated averaging and federated distillation, principled evaluation of logit- versus feature-based knowledge sharing, and bandwidth-aware feature selection to control the accuracy–communication tradeoff. By structuring these approaches under a common framework, FedBaS serves as both an analytical lens and a practical tool for studying how much knowledge must be exchanged to balance performance and bandwidth in federated learning.

VI. EXPERIMENTAL SIMULATIONS

We conduct controlled simulations to compare FedAvg with federated distillation (FedKD) variants and the proposed bandwidth-aware top- k feature selection.

A. Common Setup

Datasets. CIFAR-10 and SVHN with standard normalization; CIFAR-10 uses random crop (32×32) and horizontal flip.

Models. A lightweight CNN (3 conv blocks, GAP, linear head) with a 128-d penultimate feature, and a CIFAR-style ResNet-18 (3×3 stem, no initial max-pool). To keep ResNet-18 communication comparable to the CNN FedAvg budget (≈ 1.9 MB per direction per round), we freeze the backbone and train/broadcast only the classifier head (HO), use fp16 tensors, and choose N_{pub} so uplink ≤ 1.2 MB/round.

Federated configuration. $N=10$ clients with non-IID Dirichlet label split ($\alpha=0.5$); each round samples $m=5$ clients, trains 1 local epoch (batch 128), and updates the global model. Optimizer: SGD (lr 0.01, momentum 0.9, weight decay 5×10^{-4}). We run 10 rounds with seed 42. 10% of the training set is reserved as a public pool for distillation using a capped subset of size N_{pub} each round (sampled without replacement). We use $N_{\text{pub}}=512$ for the CNN and tune N_{pub} for ResNet-18 satisfying the uplink budget.

Accounting. We log test accuracy and uplink/downlink MB per round, reported as totals aggregated over the sampled clients ($m=5$). For FedAvg (CNN), communication is ≈ 1.9 MB per direction per round (fp32); ResNet-18 runs use fp16 and head-only broadcast to remain in a comparable range.

B. Methods Compared

FedAvg. Clients train locally and the server averages model weights (≈ 1.9 MB per direction per round for CNN, $m=5$). **FedKD.** Using the same federation and public set, we evaluate: (i) *logits-only* (KL with temperature $T=3$), (ii) *features-only* (MSE feature matching), and (iii) *logits+features* (equal loss weights). With $N_{\text{pub}}=512$ and fp16 accounting, uplink is ≤ 0.7 MB/round and downlink is ≈ 0.95 MB/round, both below the FedAvg CNN baseline.

C. Selective Feature-Only Distillation (top- k)

We evaluate a feature-only FedKD variant that transmits only a subset of feature coordinates each round. At the beginning of each round, feature dimensions are ranked, the top- k indices are selected and broadcast (negligible cost), and distillation is performed using only the selected coordinates. We consider three ranking criteria: (i) variance across the public data, (ii) classifier head weight magnitude, and (iii) estimated mutual information with labels (MINE). We sweep $k \in \{64, 128\}$ to characterize the accuracy–bandwidth trade-off. With $m=5$ clients, $N_{\text{pub}}=512$, and fp16 values, uplink scales approximately linearly with k (about 0.31 MB for $k=64$ and 0.63 MB for $k=128$), remaining well below the FedAvg baseline.

D. Evaluation Protocol and Reproducibility

All methods are evaluated for 10 rounds on both datasets using identical non-IID partitions, client sampling, local hyperparameters, and a fixed seed. Distillation uses temperature $T=3$ and equal weighting when logits and features are combined. We report test accuracy versus rounds and versus cumulative communication (MB), enabling paired and bandwidth-normalized comparisons across methods.

E. FedAvg vs FedKD

We compare FedAvg with FedKD (logits+features) on CIFAR-10 and SVHN under an identical federation setup (10 clients, 5 sampled per round, 10 rounds, same non-IID split and hyperparameters; Fig. 2). For the CNN, all layers are exchanged, while for ResNet-18 under FedKD we use a head-only configuration to limit downlink. On CIFAR-10, FedKD attains slightly lower final accuracy than FedAvg but at substantially lower communication cost: a 40.8% reduction

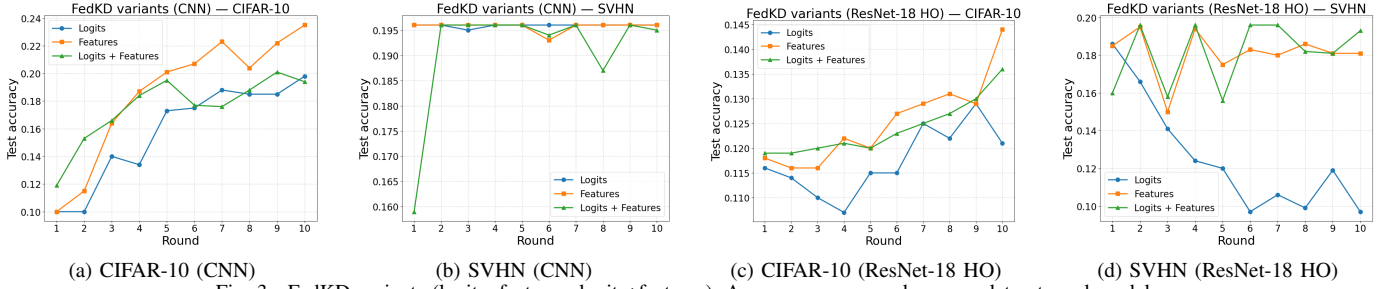


Fig. 3. FedKD variants (logits, features, logits+features). Accuracy over rounds across datasets and models.

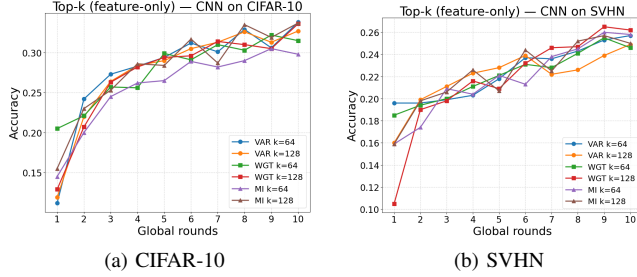


Fig. 4. FedBaS: accuracy vs. rounds for $k \in \{64, 128\}$ across VAR/WGT/MI with lightweight CNN.

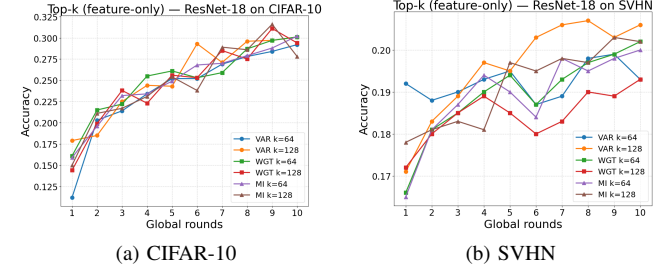


Fig. 5. FedBaS: accuracy vs. rounds for $k \in \{64, 128\}$ across VAR/WGT/MI with ResNet-18.

for the CNN and a 66.3% reduction for head-only ResNet-18. On SVHN, FedKD matches or slightly exceeds FedAvg accuracy for the CNN while using about 40% less bandwidth, and the head-only ResNet-18 variant achieves the highest accuracy-per-megabyte due to minimal downlink. Across both datasets, the accuracy-per-cumulative-MB curves consistently favor FedKD, indicating that distillation extracts more utility per transmitted byte than full-model averaging despite minor losses in absolute accuracy.

F. FedKD Variants: Logits vs Features vs Logits+Features

We compare three knowledge signals—logits only, features only, and logits+features—under identical federation and public-set budgets (Fig. 3). For the CNN on CIFAR-10, feature sharing achieves the fastest and highest improvement, while logits only converges more slowly and the combined variant performs slightly below features alone, indicating the importance of intermediate representation matching. On SVHN, all variants quickly saturate, but logits-only exhibits occasional instability, whereas feature-based variants remain smooth. For head-only ResNet-18, the benefits of feature sharing are more pronounced: logits-only degrades significantly, while features and logits+features maintain stable accuracy on both datasets. Across all settings, two consistent patterns emerge: feature sharing yields higher final accuracy and smoother trajectories than logits only, and combining logits with features is never worse than logits alone and often approaches the performance of features alone. Overall, when bandwidth permits, feature-based distillation provides more reliable convergence under the same communication budget.

G. FedBaS: Top-k Feature-Only

We make feature sharing lightweight using four practical knobs: fp16 tensors, a 512-sample public set per round, period-2 downlink with head-only broadcast, and top- k feature selection. We evaluate $k \in \{64, 128\}$ with three ranking criteria—variance (VAR), weight magnitude (WGT), and mutual information (MI)—on CIFAR-10 and SVHN using 10 clients (5 sampled per round) for 10 rounds. Under this setup, mean uplink is ≈ 0.33 MB/round for $k=64$ and ≈ 0.66 MB/round for $k=128$, corresponding to $\sim 91\%$ and $\sim 83\%$ reductions relative to FedAvg.

Across both backbones and datasets (Figs. 5, 4), top- k variants converge steadily and match or exceed FedAvg final accuracy despite partial feature sharing. On CIFAR-10, runs end in the 0.31–0.34 range (best: VAR, $k=64$), while on SVHN final accuracy lies around 0.25–0.26 (best: WGT, $k=128$). VAR generally performs best on CIFAR-10, WGT becomes competitive as k increases and leads on SVHN, and MI yields stable late-round behavior. Smaller k typically offers the best communication–accuracy trade-off, while $k=128$ provides modest gains at higher cost. Normalized by bandwidth, top- k FedBaS achieves substantially higher accuracy per MB than FedAvg (Table II), indicating that selectively transmitting informative features yields superior efficiency under tight communication budgets.

H. Future Work

The future work will be to extend FedKD to more realistic 6G edge settings that involve dynamic bandwidth, mobility, and diverse devices. We will work on developing communication strategies that adaptively choose k and periodic broadcast intervals depending on the link quality, latency, or reliability,

TABLE II
FINAL ACCURACY AND EFFICIENCY AFTER 10 ROUNDS FOR DIFFERENT VARIANTS: (FINAL ACCURACY / ACCURACY GAIN PER MB)

Dataset	Model	FedAvg	FedKD (Logits)	FedKD (Feat.)	FedKD (L+F)	FedBaS $k=64$ (VAR — WGT — MI)			FedBaS $k=128$ (VAR — WGT — MI)						
CIFAR-10	CNN	0.212 / 0.0056	0.198 / 0.0191	0.235 / 0.0110	0.194 / 0.0087	0.338/0.0142	—	0.315/0.0132	—	0.298/0.0125	0.327/0.0121	—	0.336/0.0124	—	0.337/0.0124
SVHN	CNN	0.191 / 0.0050	0.196 / 0.0189	0.196 / 0.0091	0.195 / 0.0087	0.257/0.0108	—	0.246/0.0103	—	0.258/0.0108	0.249/0.0092	—	0.262/0.0097	—	0.250/0.0092
CIFAR-10	ResNet-18	0.257 / 0.0068	0.128 / 0.1730	0.144 / 0.0115	0.136 / 0.0107	0.292/0.0826	—	0.301/0.0851	—	0.302/0.0855	0.301/0.0442	—	0.294/0.0432	—	0.278/0.0409
SVHN	ResNet-18	0.197 / 0.0052	0.097 / 0.1311	0.181 / 0.0145	0.193 / 0.0152	0.193/0.0546	—	0.202/0.0571	—	0.170/0.0481	0.206/0.0303	—	0.193/0.0284	—	0.202/0.0297

and analyze hybrid update methods that periodically update the backbone to counteract the effects of high mobility. Additional efficiency improvements can be made by integrating top- k feature selection with learned projections, structured sparsity, ultra-low precision quantization, and error feedback, and by adjusting the trade-off between logit and feature losses, and exploring multi-layer or multi-teacher distillation. Lastly, we will assess the robustness to extreme non-IID data, stragglers, and increase the number of clients and models while measuring end-to-end latency and energy cost, which are very important for vehicular and 6G edge AI systems.

VII. CONCLUSION

We studied communication-efficient federated learning by comparing FedAvg with several FedKD variants under a fixed, reproducible setup. Across CIFAR-10 and SVHN, FedKD matched or closely approached FedAvg accuracy while using substantially less bandwidth; feature-based targets consistently outperformed logits alone, and a lightweight head-only configuration kept ResNet-18 communication near the small-CNN budget. Further, a naive top- k feature selection approach leads to a significant improvement in accuracy per megabyte of data, at times outperforming FedAvg by a large margin. The above findings clearly indicate that the selective distillation and sharing of informative features can serve as a viable alternative to model sharing for federated learning in communication-limited edge and 6G settings. The proposed methods (fp16 tensors, public set truncation, periodic head updates, and top- k feature sharing) are easy to implement and can be combined with existing optimizations.

REFERENCES

- [1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y. Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2017.
- [2] E. Jeong, S. Oh, H. Kim, J. Park, M. Bennis, and S.-L. Kim, "Communication-efficient on-device machine learning: Federated distillation and augmentation under non-iid private data," in *Proceedings of the 32nd Conference on Neural Information Processing Systems (NeurIPS) Workshop*, 2018.
- [3] S. Yang, X. Wang, Y. Lin, T. Han, Q. Xu, and Y. Xu, "Distillfl: Communication-efficient federated learning by adaptive knowledge distillation," *IEEE Transactions on Neural Networks and Learning Systems*, 2021.
- [4] T. Lin, L. Kong, S. U. Stich, and M. Jaggi, "Ensemble distillation for robust model fusion in federated learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [5] A. Romero, N. Ballas, S. E. Kahou, A. Chassang, C. Gatta, and Y. Bengio, "Fitnets: Hints for thin deep nets," *arXiv preprint arXiv:1412.6550*, 2015.
- [6] D. Alistarh, D. Grubic, J. Li, R. Tomioka, and M. Vojnovic, "Qsgd: Communication-efficient SGD via gradient quantization and encoding," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 1709–1720.
- [7] W. Wen, C. Xu, F. Yan, C. Wu, Y. Wang, Y. Chen, and H. Li, "Terograd: Ternary gradients to reduce communication in distributed deep learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 1509–1519.
- [8] A. F. Aji and K. Heafield, "Sparse communication for distributed gradient descent," in *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2017, pp. 440–445.
- [9] Y. Lin, S. Han, H. Mao, Y. Wang, and W. J. Dally, "Deep gradient compression: Reducing the communication bandwidth for distributed training," in *International Conference on Learning Representations (ICLR)*, 2018.
- [10] J. Konečný, B. McMahan, D. Ramage, and P. Richtárik, "Federated optimization: Distributed machine learning for on-device intelligence," in *arXiv preprint arXiv:1610.02527*, 2016.
- [11] S. P. Karimireddy, S. Kale, M. Mohri, S. Reddi, S. Stich, and A. Suresh, "Scaffold: Stochastic controlled averaging for federated learning," in *International Conference on Machine Learning (ICML)*, 2020, pp. 5132–5143.
- [12] A. Reiszadeh, F. Farnia, R. Pedarsani, and A. Jadbabaie, "Fedpaq: A communication-efficient federated learning method with periodic averaging and quantization," in *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2020, pp. 2021–2031.
- [13] S. Zagoruyko and N. Komodakis, "Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer," in *International Conference on Learning Representations (ICLR)*, 2017.
- [14] W. Park, D. Kim, Y. Lu, and M. Cho, "Relational knowledge distillation," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 3967–3976.
- [15] D. Li, J. He, and A. Song, "Fedmd: Heterogenous federated learning via model distillation," in *arXiv preprint arXiv:1910.03581*, 2019.
- [16] T. Lin, L. Kong, S. U. Stich, and M. Jaggi, "Ensemble distillation for robust model fusion in federated learning," in *Advances in Neural Information Processing Systems (NeurIPS) Workshop*, 2020.
- [17] R. Battiti, "Using mutual information for selecting features in supervised neural net learning," *IEEE Transactions on Neural Networks*, vol. 5, no. 4, pp. 537–550, 1994.
- [18] H. Peng, F. Long, and C. Ding, "Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 27, no. 8, pp. 1226–1238, 2005.
- [19] N. Tishby, F. Pereira, and W. Bialek, "The information bottleneck method," in *Proceedings of the 37th Annual Allerton Conference on Communication, Control, and Computing*, 2000, pp. 368–377.
- [20] A. Alemi, I. Fischer, J. Dillon, and K. Murphy, "Deep variational information bottleneck," in *International Conference on Learning Representations (ICLR)*, 2017.
- [21] T. Li, A. K. Sahu, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proceedings of Machine Learning and Systems (MLSys)*, 2020.
- [22] M. G. Arivazhagan, R. Aggarwal, A. Singh, and S. Choudhary, "Federated learning with personalization layers," in *arXiv preprint arXiv:1912.00818*, 2019.
- [23] A. Fallah, A. Mokhtari, and A. Ozdaglar, "Personalized federated learning with theoretical guarantees: A model-agnostic meta-learning approach," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.